

A. HARSHA GANESH

Hardware & PCB Design Engineer | Embedded Systems

Tirupati, Andhra Pradesh, India 517502 | +91-9844553752

harshaganesh300@gmail.com | linkedin.com/in/harsha-ganesh-ee | github.com/Harshavar8jdm

Summary

Final-year Electronics and Communication Engineering undergraduate (CGPA 8.8) with proven expertise in end-to-end PCB design, board bring-up, and hardware debugging. Designed, routed, and bench-tested 17+ custom multi-layer and power PCBs using KiCad, spanning power converters, BMS, and high-speed microcontroller boards. Demonstrated proficiency in schematic capture, component selection, board bring-up, and signal integrity troubleshooting using oscilloscopes and logic analyzers. Validated through internships at IIT Palakkad, IIT Tirupati, and Hypstuma, with core skills in C, Python, Verilog, and circuit-level failure analysis. Seeking an entry-level PCB / Hardware Design Engineer role.

Education

Sri Venkateswara College of Engineering	2023 – 2027
<i>B.Tech, Electronics and Communications Engineering</i>	CGPA: 8.8
Gnanadhare PU College	2021 – 2023
<i>Intermediate</i>	88.8% (532/600)

Experience

IIT Palakkad	May 2026 – June 2026
<i>Digital Systems & Real-Time Simulation Intern</i>	Palakkad, India

- Implemented a real-time mathematical model of a PMDC motor on the programmable-logic (PL) side of a Zybo Z7-20 (Zynq) FPGA board at a $1\mu\text{s}$ sampling rate, streaming processed samples to a PC over Gigabit Ethernet and visualizing the motor's transient and steady-state response in a Python (Matplotlib/Tkinter) front-end.

Hypstuma PVT LTD	July 2025 – August 2025
<i>Drone Design Intern</i>	Chennai, India

- Optimized power distribution and signal integrity for multi-rotor UAV platforms using Pixhawk 6C flight controllers and 40A BLHeli_32 ESCs, ensuring stable communication with M8N GPS modules.
- Performed PID tuning to reduce oscillation by 15% and validated 2.4GHz ELRS telemetry links for sub-50ms latency – iterative hardware bring-up, calibration, and fault diagnosis on live flight hardware.

IIT Tirupati Navishkar I-Hub Foundation	June 2025 – July 2025
<i>GPR Simulation Intern</i>	Tirupati, India

- Automated parameter variations (depth, size, material) to generate a dataset of 23,000 parametric ground-penetrating-radar simulations for machine learning applications, built around a custom PyQt5 tool.

PCB Design Project Portfolio (KiCad)

Full designs on GitHub: github.com/Harshavar8jdm

Power Electronics – Converters & Chargers: LM2596S buck converter; LMR51450 buck converter; LMR16006 12V-to-3.3V buck converter; TPS61040DBVR 5V-to-12V boost converter; BQ24092DGQT 1S Li-ion charger PCB; USB-PD triggered power supply module.

Battery Management Systems: HY2112-based 1S LiFePO4 BMS PCB; BQ79150WPWT 5S BMS for NiMH battery packs; BME280 environmental sensor breakout module.

Microcontroller & Interface Boards: ESP32-S3 development board; STM32F401RCT6 development board; CP2102 USB-to-UART bridge board; micro-SD card breakout board.

Test, Measurement & Automotive Electronics: Custom 2-channel oscilloscope built around an STM32 MCU; IoT-based real-time power quality analyzer (ESP32 + ACS712) with isolated AC/DC layout for THD monitoring; custom engine monitoring unit; capacitor-discharge ignition (CDI) module for single-cylinder motorcycles.

Technical Skills

PCB Design & EDA: Schematic capture, multilayer PCB layout, Gerber generation, and signal-integrity optimization in KiCad and EasyEDA; power-electronics and mixed-signal (AC/DC isolation) layout.

Hardware Bring-Up & Debugging: Board-level functional testing, multimeter/oscilloscope-based fault tracing, continuity and power-rail checks, register-level firmware debugging.

Microcontrollers & Digital Systems: STM32, ESP32, ATmega328P, RP2040 (Raspberry Pi Pico), Arduino, Zynq/FPGA (Zybo Z7-20); bare-metal C programming, RTOS principles.

Programming & Frameworks: C, Python (PyQt5, Selenium, Streamlit), Java, MATLAB/Simulink, Verilog.

Systems & Flight Hardware: Pixhawk 6C, M8N GPS, ELRS telemetry.